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In an effort to address Mr. Kiser's needs, our team compiled a list of considerations including the product's safety, manufacturability, adjustability, comfort, aesthetic appeal, and longevity. All of these considerations are critical factors that contribute to the overall viability of the design.

Safety and Manufacturability:

Our design aligns with the principles of safety and manufacturability, which are foundational to ensuring that the handlebar extension is both reliable and feasible for mass production.

Throughout the design process, we focused on using materials and construction methods that are durable and safe for everyday use. For example, we used PVC pipes to ensure that the edges of the handle are rounded to prevent discomfort or injury -- a key consideration for ergonomics.

Additionally, we selected materials that are easy to source and work with, such as PVC and wood, to keep the manufacturing process cost-effective and straightforward due to extreme cost-restraint. The use of these materials was also verified through expert feedback, ensuring that the final product can be reliably produced at scale without compromising on safety.

Adjustability and Comfort:

The design's adjustability is a crucial feature, as Mr. Kiser's primary concern was alleviating back strain caused by bending over when pushing the cart; however, we could not exclude those of shorter heights -- a demographic filled by a couple members of the team. By incorporating an adjustable handle length, this design can tailor the product to Mr. Kiser's height and specific needs. The adjustment mechanism, which relies on a halved PVC pipe and adjustable velcro, allows for a secure fit while remaining easy to modify depending on the load and user preferences. Vanderbilt expert feedback emphasized the importance of comfort, particularly with

regards to the grip to the original handle. The team applied suggestions to add soft grips, using materials like pool noodles, caulk, or plastic tubing, which were proven to enhance the ergonomics of the handle, ensuring that the product is comfortable for more aggressive use.

Aesthetic Appeal:

While functionality was the primary focus, aesthetic considerations were not overlooked. The design avoids overly utilitarian or unattractive elements, and attention was paid to ensuring that the handlebar extension looks polished and visually consistent with the cart. The addition of plastic tubing for grip, and the recommendation to replace duct tape with more permanent and visually appealing solutions, demonstrates that the design accounts for the product's appearance without compromising on performance. This attention to detail ensures that the handlebar extension will be both functional and aesthetically pleasing, in line with Mr. Kiser's desire for a product that looks good as well as perform well.

Longevity:

Another crucial extra-functional consideration was the product's durability. Given that Mr. Kiser requires a solution that will last for at least five years, we focused on selecting materials and design features that would stand the test of time. The PVC pipes, wood, and plastic tubing are all durable and able to withstand regular use without degrading or losing functionality. Moreover, the design's ability to adjust to various cart sizes and loads means that the handlebar extension will remain effective as Mr. Kiser's needs evolve over time. The advice from engineers about reducing rigidity and allowing mobility ensures that the handle will continue to function well even after prolonged use.

Viability of the Design:

Based on these extra-functional considerations, the proposed handlebar extension design is not only viable but also well-supported by credible evidence. The expert input from Mrs. Piper and Mr. Jared provided invaluable insights into how the design could be improved to better address all of the key considerations—comfort, safety, aesthetics, and durability—while ensuring ease of use and manufacturability. Additionally, by continually refining the prototype with feedback from experienced engineers, we are confident that the final solution will effectively address Mr. Kiser's needs without compromising on quality or usability. The balance between practicality, safety, and aesthetics, combined with the team's adherence to proven engineering principles, ensures that the proposed design is realistic and capable of delivering long-term value to Mr. Kiser.

